Chem 456 1/12/2026

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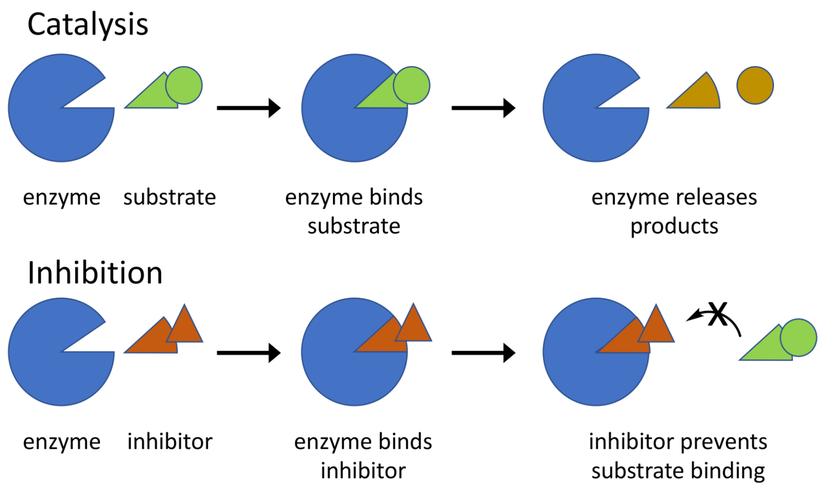
Computational Biochemistry and Drug Design

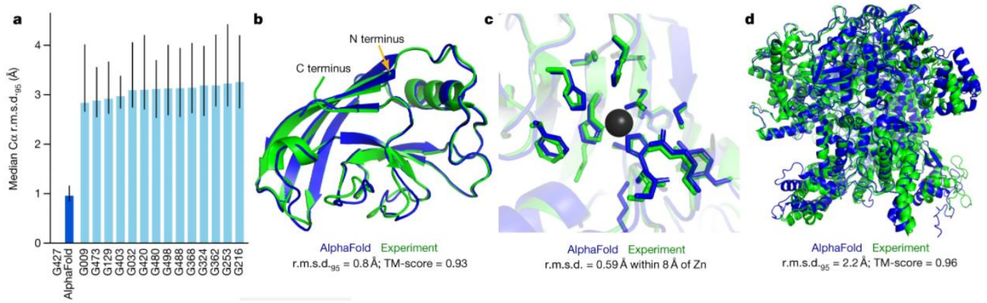
- **Biochemistry** – the study of chemical processes related to living organisms.
- **Drug design** – the inventive process of finding new medications based on the knowledge of a biological target.
- Nowadays, computers are widely used in both.



Some biochemical processes. CCA-SA4.0

What will this course cover?

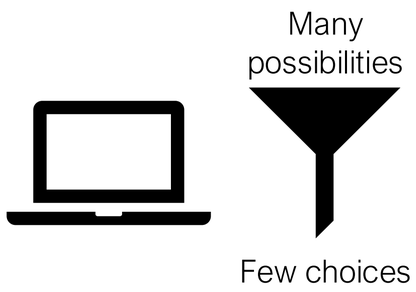
- **Informatics** – the use of data-centric approaches to solve complex problems. Includes artificial intelligence.
- **Energy**, focusing on types and applications of potential energy models
- **Molecular dynamics simulations**
  - evaluating stability of structural elements and networks
  - investigating how motion leads to conformational populations and relates with function



AlphaFold protein structure prediction (Nature 596, 583–589 (2021) CCA-SA4.0) was awarded the 2024 Nobel Prize in Chemistry

Some tools we'll learn to use

- AlphaFold
  - Informatics-based structure prediction
  - Less track record for complexes
- Molecular docking
  - Physics-based structure prediction
  - Rank (large) virtual libraries to prioritize molecules to test
- Enhanced sampling methods
  - Explore conformations more quickly
  - Obtain reproducible and converged results
- Free energy methods
  - Compare interactions between (closely analogous) molecules
  - Rank (small) virtual libraries more rigorously



Milestones related to Chem 456

2004 — Molecular modeling workshop
I attended a workshop at the start of my Ph.D., helping me get off to a quick start

2017 — Programs launched
Illinois Tech launches degree programs in medicinal and computational chemistry

2020 — Chem 456 first taught
Chem 456 begins in a computer lab and moves online due to the pandemic

2021M — Online workshop + Chem 456
I organized an online workshop and ran a small session of Chem 456, running labs on a virtual machine

2022S — Colombia workshop
I worked with four students to run a workshop in Colombia, migrating labs to Google Colab

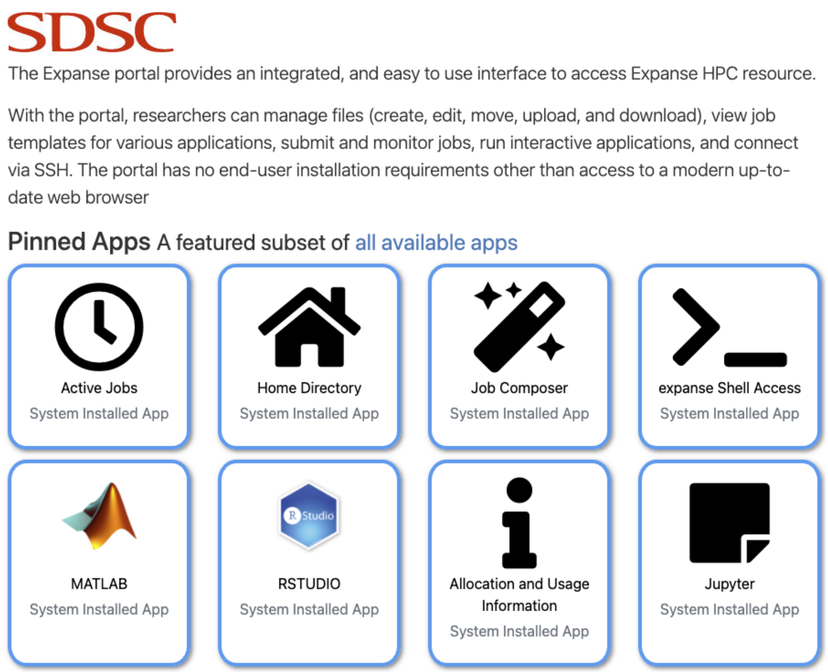
2022F — Patent + startup outcome
I taught Chem 456 and a student project ultimately led to a patent and a startup company

2024F — Nobel Prize in Chemistry
Awarded for “computational protein design” and “protein structure prediction”, related to this class

2024F — ACCESS-CI migration
I taught Chem 456 and migrated some labs to ACCESS-CI

Online Resources

- The course website includes the
  - Syllabus
  - Resources tab, with links to previous classes and other sites
  - Schedule, with links to class notes and exercises
  - Description of Assignments and Grading
- The San Diego Supercomputer Center (SDSC) Expanse User Portal
  - requires an account from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS-CI) program
  - can be accessed through modern laptop/desktop computers (e.g. classroom desktops) using a modern browser
  - allows you to
    - upload and download files
    - access the shell to set up environments/submit batch jobs and run interactive Jupyter notebooks
  - is where you will work on exercises and projects
  - will use an educational allocation from ACCESS-CI. If needed, graduate students can obtain their own allocation.
- Canvas is where you will
  - submit assignments, except for large files
  - post information in discussions visible to the class



Introductions

About me: work

Biographical

Professional History

2020 – present	**Robert E. Frey, Jr. Endowed Faculty in Chemistry**, Illinois Institute of Technology (Illinois Tech), Chicago, IL
2019 – present	**Associate Professor** (tenured), Department of Chemistry, Illinois Tech
2018 – present	**Associate Director**, Center for Interdisciplinary Scientific Computation, Illinois Tech
2013 – 2019	**Assistant Professor** (tenure-track), Department of Chemistry, Illinois Tech
2011 – 2013	**Postdoctoral Research Associate**, Duke University, Durham, NC
2009 – 2011	**Director's Postdoctoral Fellow**, Argonne National Laboratory, Argonne, IL
2007 – 2009	**Postdoctoral Trainee**, National Institutes of Health, Bethesda, MD

Education

2004 – 2007	**Ph.D. in Physical Chemistry**, University of California, San Diego. Thesis Title: Free Energy Reconstruction from Irreversible Single-Molecule Pulling Experiments. Recipient of Molecular Biophysics Training Grant and Aguoron Kamen and Kaplan Fellowship.
2000 – 2003	**B.A. in Chemistry**, University of California, Berkeley. Recipient of Chancellor's Scholarship (Berkeley's most prestigious scholarship) and National Merit Scholarship.

Awards

2020	College of Letters and Science Nominee, Michael J. Graf IIT Teaching and Advising Innovation Award.
2019	40 under 40 Chicago Scientists. Selected by Halo Cures.
2019	Sigma Xi Junior Faculty Award, in recognition of Outstanding Accomplishments in Research and Scholarship. Awarded by the Illinois Tech chapter of the scientific research honor society, Sigma Xi.
2018	College of Science Dean's Excellence Award in Research, at the Junior Level
2012	OpenMM Visiting Scholar (at Stanford)
2009 – 2011	Director's Postdoctoral Fellowship
2007 – 2009	Postdoctoral Intramural Research Training Award
2005 – 2007	NIH Molecular Biophysics Training Grant
2004 – 2005	Aguoron Kamen and Kaplan Fellowship
2000 – 2003	UC Berkeley Chancellor's Scholarship

- Coauthored 61 peer-reviewed journal articles
- Cited over 1700 times with an h-index of 23, according to Google scholar.
- Current research in computational chemistry
  - Developing methods related to structure-based drug design
    - predicting binding affinities and ligand efficacy
    - simulating biological macromolecules
  - Involved in antibiotic drug discovery
  - Research has been funded by NIH, NSF, and Illinois

Full curriculum vitae: https://ccbatiit.github.io/downloads/DavidMinh_CV.pdf

About me: beyond work

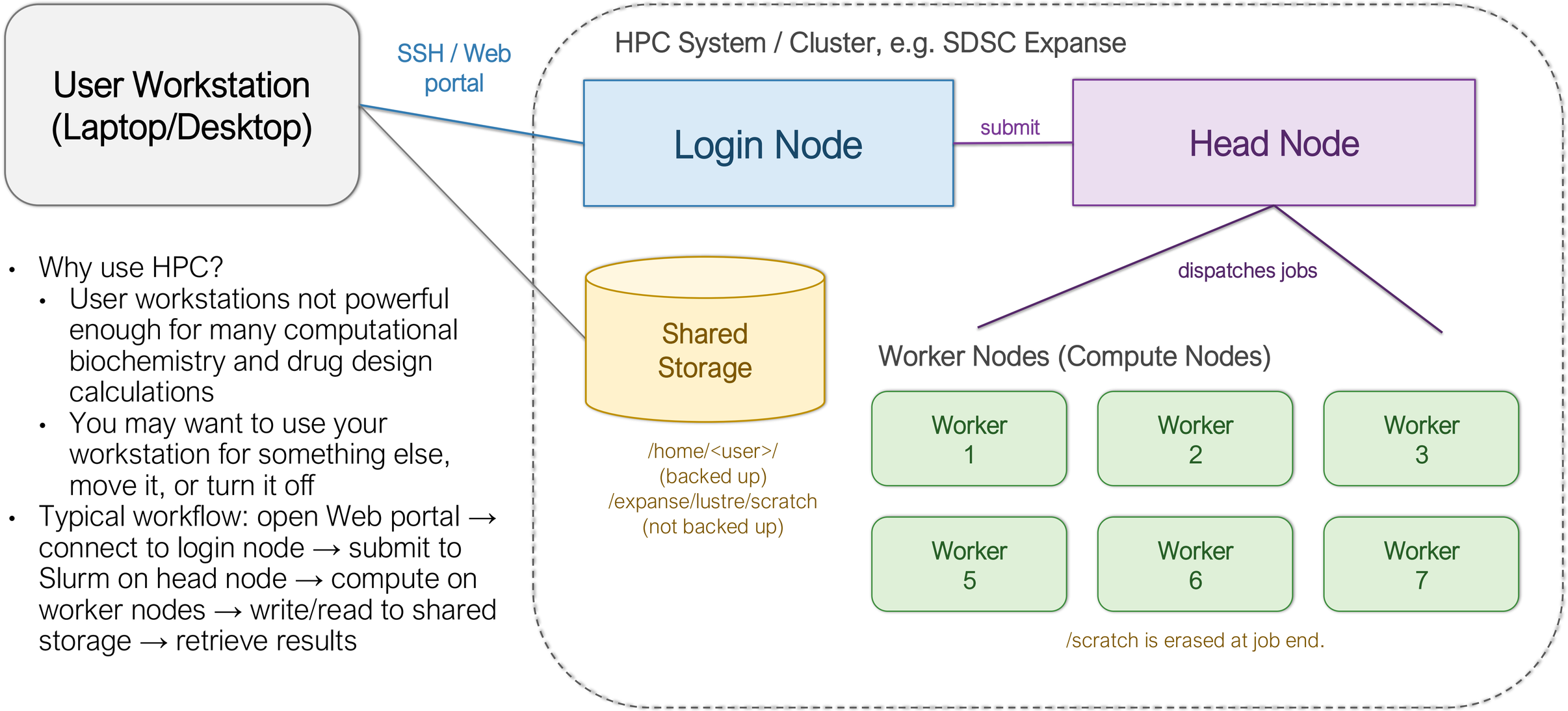
- Outside of work, I like to
  - exercise, especially CrossFit, swimming, and biking.
  - play sports, especially basketball, tennis, and table tennis.
  - listen to podcasts and sometimes audiobooks. Some of my favorite podcasts are Planet Money and Hidden Brain.
  - play board games, especially chess.
  - play music, especially guitar. I also can play bass and piano.
  - read news and sometimes books.
  - travel. I've been to 6 continents.
- I am
  - Australian-born and American-raised by Vietnamese parents of Vietnamese and Chinese descent
  - married, with 2 children
  - Christian, c.f. my profile on Meet the Prof
  - incredibly blessed!

Introduce yourself

- Before the next class, fill out Meet Your Classmates on Canvas
  - What is your full name? What do you like to be called?
  - Which degree program are you in?
  - What is one things that you hope to learn in this class?
  - How will this help you achieve your goals?
  - Share something interesting about yourself

High performance computing

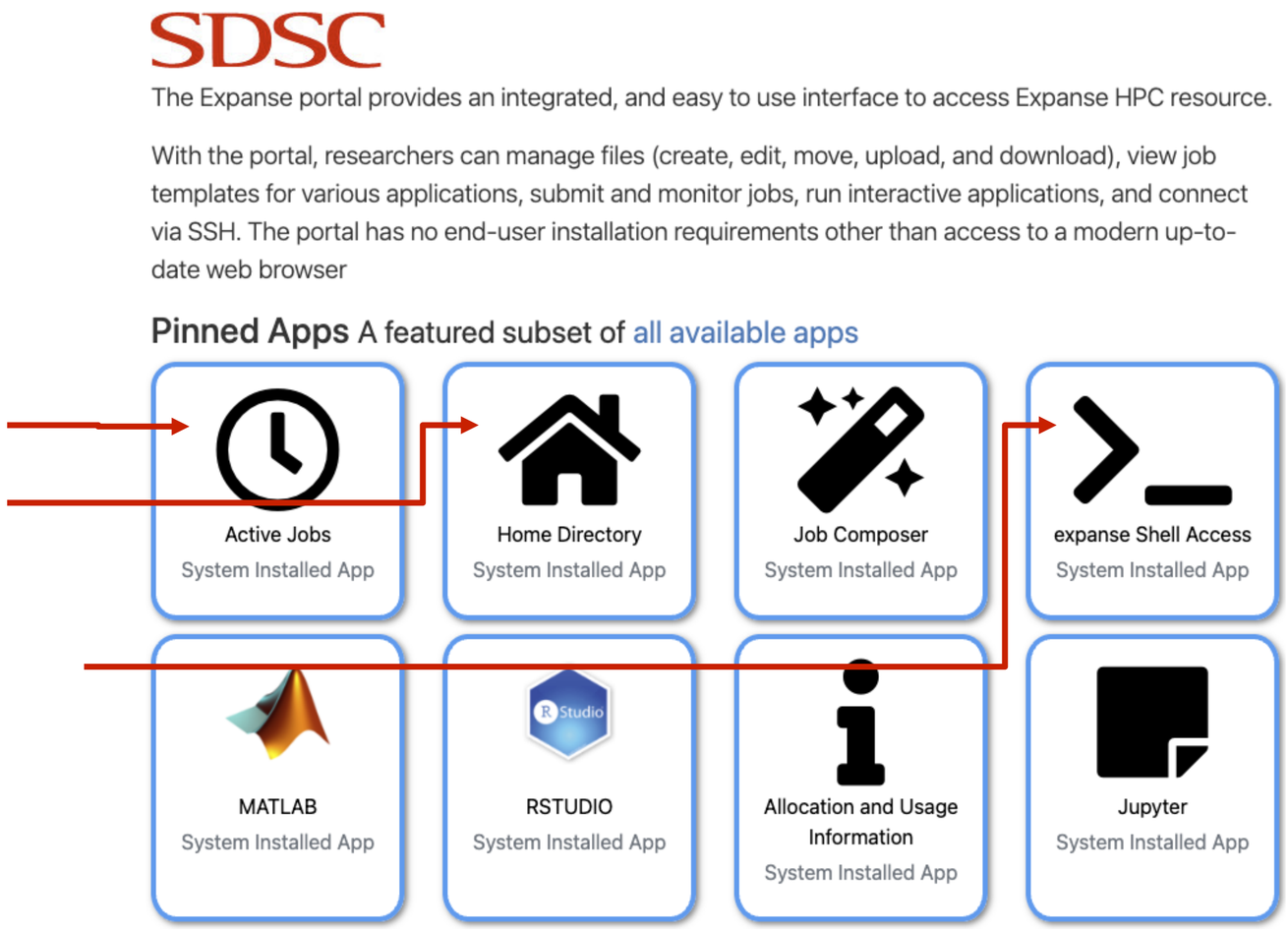
High Performance Computing (HPC) Systems



- Why use HPC?
  - User workstations not powerful enough for many computational biochemistry and drug design calculations
  - You may want to use your workstation for something else, move it, or turn it off
- Typical workflow: open Web portal → connect to login node → submit to Slurm on head node → compute on worker nodes → write/read to shared storage → retrieve results

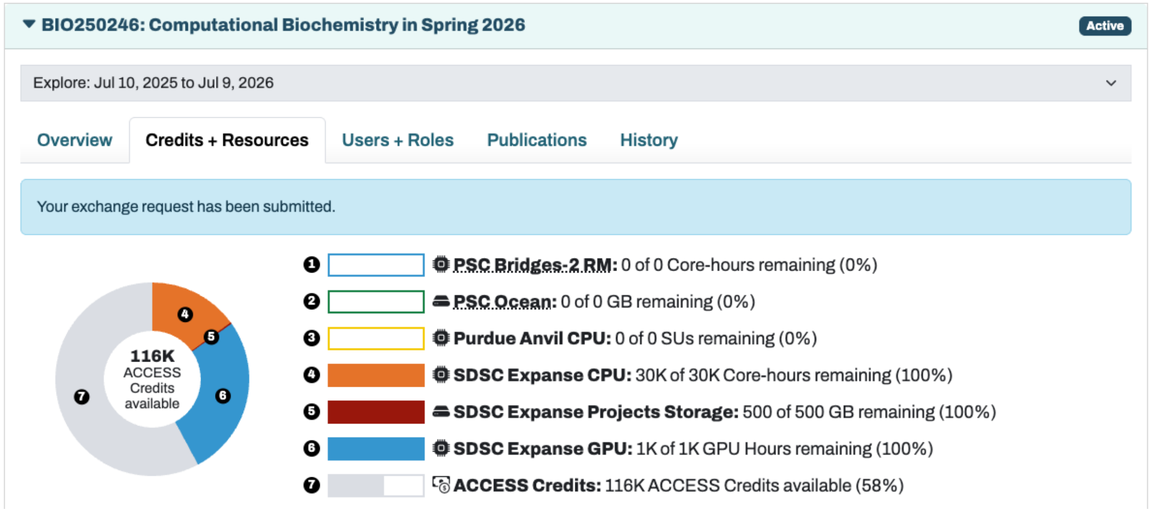
The SDSC Expanse User Portal

- Logging into the User Portal requires an ACCESS-CI account
- In this class we will mostly
  - manage jobs (close when done)
  - upload and download files
  - access the shell to set up environments, submit jobs, and run interactive Jupyter notebooks



ACCESS Allocations

- System for distributing ACCESS computing resources
- Class (not individual students) has one allocation
- Graduate students / PhD scientists can be PIs on additional ACCESS allocations
- Allocation in ACCESS Credits that can be exchanged for resources, mostly CPU, GPU, and Storage
- c.f. https://allocations.access-ci.org/



Basic Linux

Why Linux?

- Linux is a practical “work environment” for doing computational science — especially when you move beyond small, local runs and start using HPC
  - Open source + customizability for specialized hardware and workloads
  - Cost-effective clustering with commodity nodes
  - Strong job scheduling + cluster management ecosystem
  - First-class support for parallel computing standards (MPI)
  - Operational reliability for long-running, multi-user systems
- We will use the Linux terminal to
  - Manage files and environments on ACCESS-CI HPC resources
  - Start up interactive course notebooks
  - Submit & monitor longer calculations as batch jobs

The Shell (bash)

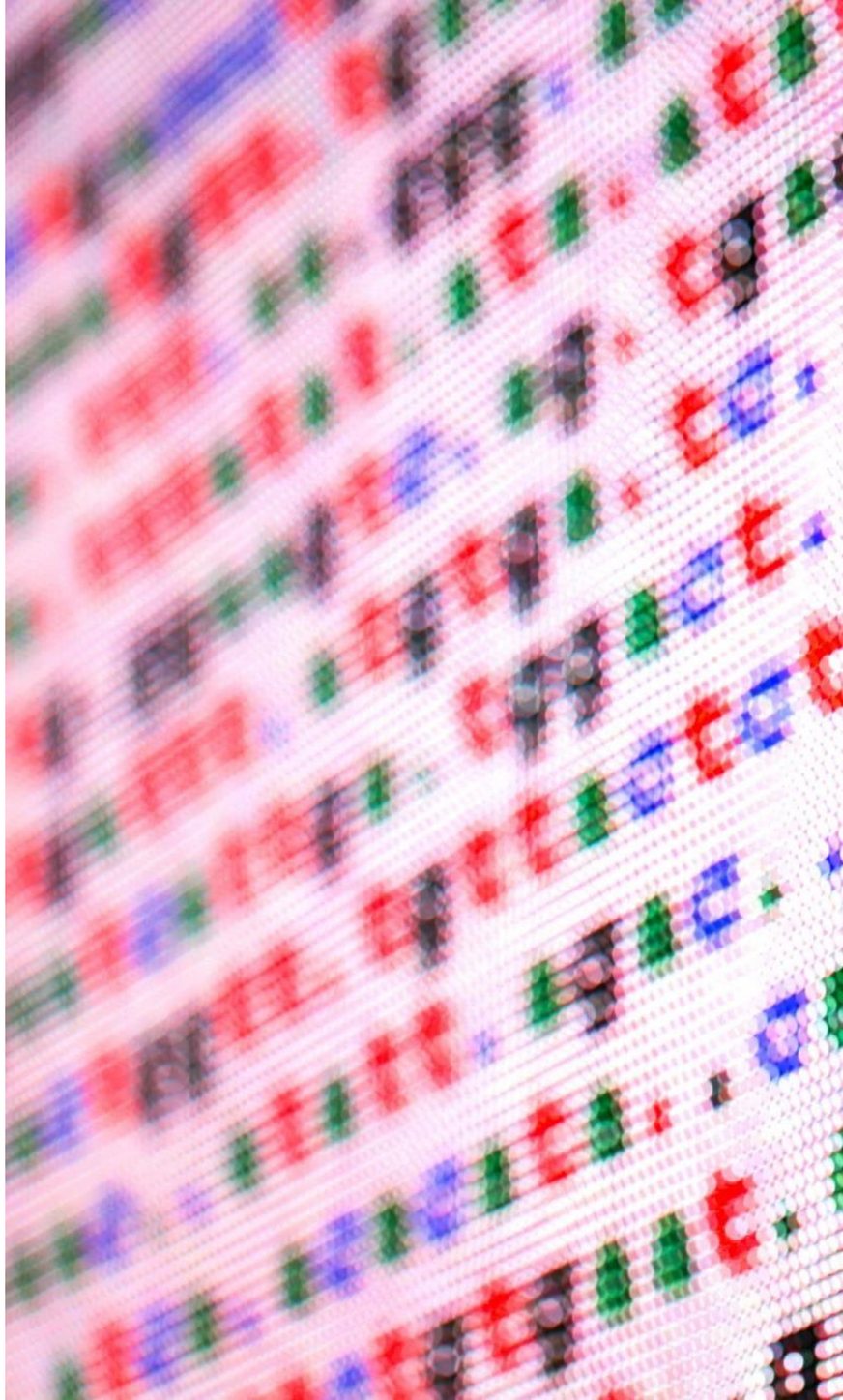
Core Shell Commands
Commands like 'pwd', 'ls', and 'cd' enable navigation and file listing in the shell environment.

File Operations
'mkdir', 'cp', 'mv', and 'rm' manage directories and files.

We will practice basic commands and file operations in Exercise 0.

Advanced Shell Features (Optional)
Redirection and piping allow chaining commands and controlling input/output streams efficiently.

Help and Automation
Commands like 'man' and '--help' assist users; shell scripts automate batch jobs on HPC clusters.



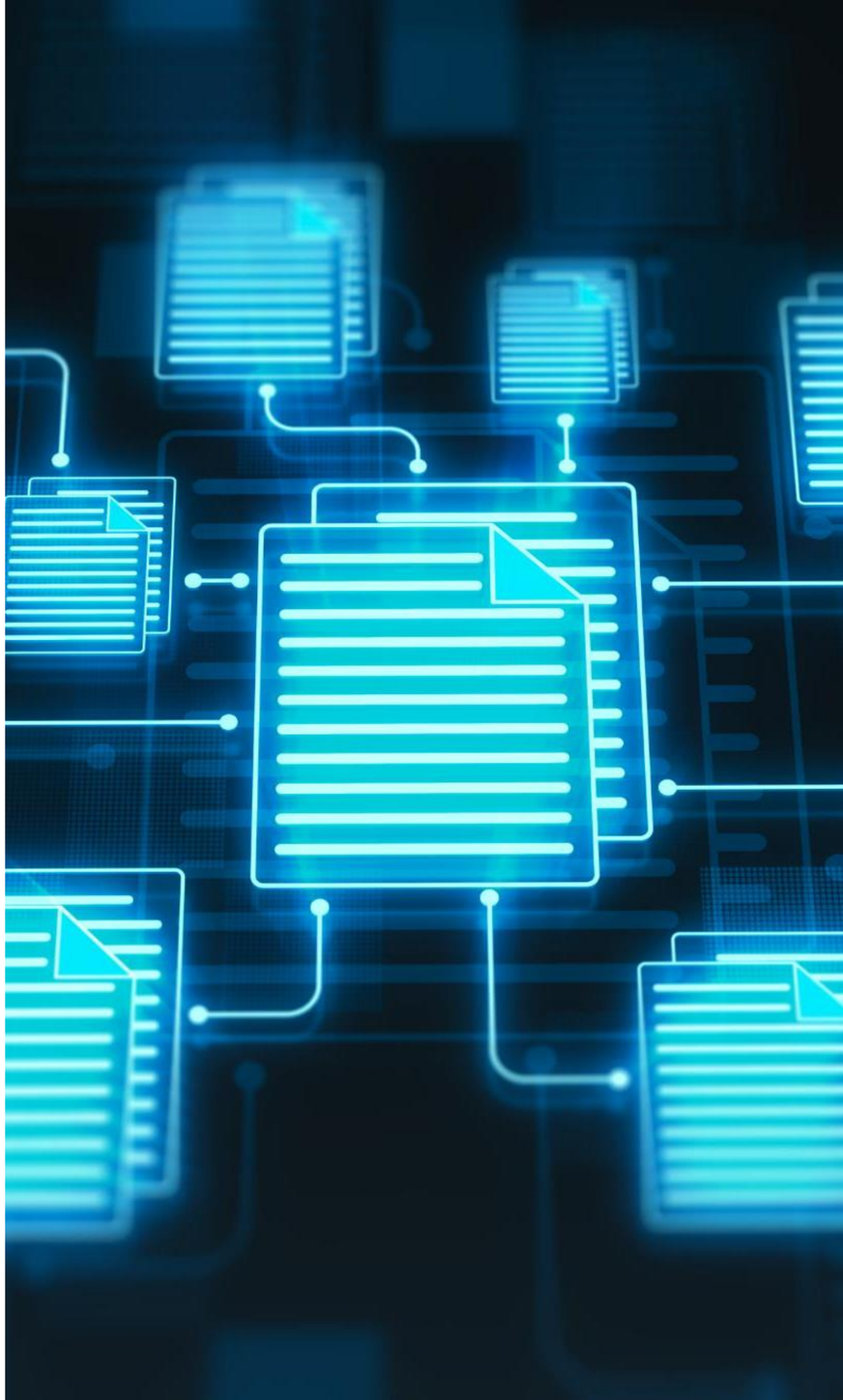
Filesystems and Paths

Hierarchical Directory Structure
Linux uses a root-based hierarchical directory tree to organize files with absolute and relative paths.

Storage Areas in HPC
HPC environments offer home directories, project spaces, and scratch spaces for different data storage needs.

Disk Space and Usage Commands
Commands like 'df -h' and 'du -sh *' help users monitor disk space and directory sizes efficiently.

File Permissions and Data Management
Understanding permissions and data locations is vital to avoid data loss and maintain storage policy compliance.



Absolute vs. Relative Paths

Absolute path = exact address

- Starts at the root: /
- Works from anywhere (independent of your current folder)
- Example: /home/$USER/chem456/project1/data/input.csv
- Analogy: an exact street address
- Shortcut: ~/ is your home directory

Relative path = directions from here

- Starts from your current directory (pwd)
- Shorter, but depends on where you are
- Example: data/input.csv (when you're in ~/chem456/project1)
- Analogy: "across the street from Pritzker Science Center"
- Shortcut: ../ is the parent directory

Permissions

Linux File Permissions
Permissions indicate who can read, write, or execute files, shown as symbols like '-rwxr-x---'.

Permission Modification
Use commands like 'chmod +x script.sh' to change file permissions and make scripts executable.

Security and Collaboration
Proper permission management prevents unauthorized access and supports group-based collaboration in HPC environments.

Common Permission Errors
Misconfigured permissions can cause errors like 'Permission denied,' disrupting workflows and operations.

Exercise 0: Basic Linux + Treasure Hunt

Homework

- By Tuesday 1/13 at noon, sign up for an ACCESS account and submit your username on Canvas. This will give ACCESS and I time to add you to the class allocation by Wednesday's class.
- By Wednesday 1/14 at 8:30am, fill out Meet Your Classmates on Canvas. This counts as two participation points.
- By next Wednesday 1/21 at 8:30am, complete Exercise 0.
- By next Friday 1/21 at 8:30am, complete Exercise 1.